

SEIF ELDEN MOSTAFA ELDABY

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EDUCATION

Assiut University Assiut, Egypt 2024 – 2029

Bachelor of Engineering, Electrical & Electronics Engineering

- GPA: Very Good (High)

EXPERIENCE

Magnum Egypt Assiut, Egypt Aug 2022 – Feb 2026

Curriculum Developer & Lead Instructor (Jan 2025 – Feb 2026)

- Developed and delivered technical curricula for Arduino and electronics courses to 20+ students per cohort.
- Designed hands-on labs and assessments; conducted online and in-person sessions.

PCB Assembly & Repair Technician (Aug 2022 – Feb 2026)

- Assembled and soldered 90+ high-precision PCBs across multiple large-scale production projects.
- Diagnosed and repaired faulty electrical components, maintaining zero-defect standards.

Technical Instructor & Assistant (Jan 2023 – Jan 2025)

- Supported lead instructors in delivering technical workshops; supervised and evaluated student work.

Idea Space Assiut, Egypt 2021 – 2022

Technical Support & Instructor Assistant (2022)

- Manufactured, soldered, and assembled PCBs; maintained and repaired electrical components.
- Supervised students' technical work during training sessions and competitions.

Volunteer Technical Assistant (2021)

- Assembled ~50 PCBs and prepared components for project-based training programs.
- Co-supervised two national-level technical competitions (2021, 2022).

LEADERSHIP & ACTIVITIES

Assiut Motorsport Assiut, Egypt Mar 2025 – Present

Autonomous Systems Engineer (Dec 2025 – Present)

- Achieved 7th place globally out of 73 international teams at RoboRacer Sim Racing League — ICRA 2026 in the team's debut year.
- Collaborated in building a full-stack autonomous vehicle software node structure from scratch using ROS 2 and Docker.
- Implemented Mapping & Localization pipeline in simulation using SLAM, Particle Filter, and LiDAR; integrated rf2o for LiDAR-based odometry within a ROS 2 / Docker environment.

Team Leader, Kickdown Competition (Sep – Nov 2025)

- Managed a multidisciplinary engineering team; assigned tasks, tracked milestones, and delivered presentations at internal reviews.

Engine Division Member (Mar 2025 – Jan 2026)

- Contributed to Formula Student car development within the Engine Division.

Astro Assiut, Egypt Sep 2024 – Present

Vice President & Head of Operations (Dec 2024 – Present)

- Oversee strategic planning and day-to-day operations for the organization.

SolidWorks Instructor (Nov 2024 – Mar 2025)

- Taught CAD design using SolidWorks to organization members; developed training materials.

PROJECTS

Custom ESP32 Development Board — *Personal Project*

- Designed and manufactured a custom ESP32-based PCB from schematic to final board using EasyEDA; included schematic design, component selection, PCB routing, and SMD soldering.

Autonomous Mapping & Localization — RoboRacer Sim Racing League, ICRA 2026

- Developed a Mapping & Localization pipeline in simulation using SLAM, Particle Filter, and LiDAR odometry (rf2o) within a ROS 2 / Docker environment.

LED Driver Circuit — AC 220V to LED — University Course Project

- Designed a transformer-based linear LED driver with bridge rectifier and series-parallel LED topology; simulated in LTSpice and produced full academic report.

PCB Projects at Magnum Egypt

- Led assembly of 90+ high-precision PCBs including smart wall switch and embedded systems boards.

Fashion-MNIST CNN Image Classifier — Personal Project (*TensorFlow, Keras, OpenCV*)

- Designed and trained a CNN to classify 10 clothing categories achieving ~91% test accuracy over 10 epochs.

SKILLS

Hardware & PCB: KiCAD, Altium Designer, EasyEDA, PCB Assembly & Soldering, Circuit Prototyping, Electrical Troubleshooting, Arduino, Embedded Systems (ESP32)

Software & Tools: LTSpice, SolidWorks (CSWA), AutoCAD, MATLAB, C++, Python, OpenCV, ROS 2, Docker, Git

Technical Domains: Analog Circuit Design, Power Electronics, Computer Vision (Classical Image Processing), Schematic Design, Autonomous Systems

Languages: Arabic (Native), English (Professional Working Proficiency)

CERTIFICATIONS & TRAINING

CSWA — SolidWorks CAD Design Associate — *Dassault Systèmes* Aug 2025

Become a Professional PCB Designer with Altium — *Udemy* Mar 2026

PCB Design Level 1 — KiCAD — *INNOVA* Feb 2026

PCB Design Introductory Workshop — EasyEDA / ESP32 — *IEEE PPU Student Branch* May 2026

Computer Vision Level 1 — Classical Image Processing — *INNOVA* Feb 2026

InnovEgypt Program (45 hrs + Startup Project) — *ITIDA / TIEC* 2026

AI for Beginners — *Almentor* Jun 2025

Basics of AI & ChatGPT Technology — *Almentor* Jun 2025

AWS Educate DECI Summer Program — *AWS + MCIT* Jul 2023

DECI-GEEKS L2 Midyear Competition + Microsoft Ideation Camp — *MCIT + Microsoft* Feb 2023

Digital Egypt Cubs Initiative — Level 2 Nanodegree — *MCIT + Udacity* Oct 2023

AWARDS & HONORS

- 7th Place Globally (73 international teams) — RoboRacer Sim Racing League, ICRA 2026 | Assiut Motorsport Autonomous Team, debut year (2026)
- Participation — AAKRUTI Innovation Competition 2025, Dassault Systèmes / SolidWorks Worldwide Product Design Contest
- GPA: Very Good (High) — Assiut University, First Year (2024–2025)